Politecnico di Torino

Master degree in Electronics Engineering

PROJECT REPORT

Electronic System Engineering

BarcOdewarehouse



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Project specifications, features and ratings

1.1 Project Specification

The objective of this project was to develop a Warehouse Barcode Scanner utilizing advanced technology for efficient inventory management. The key component chosen for this application is the AR0130CSSC00SPBA0 CMOS image sensor, known for its high-resolution imaging capabilities. The system is designed to scan and decode barcodes swiftly to streamline warehouse operations.

1.2 Features and Ratings

1. High-resolution imaging capabilities
2. Fast and accurate barcode scanning
3. Integration of the Texas Instruments CC2640R2LRHBR microcontroller for efficient data processing
4. Utilization of the MCP73833 USB battery charger for convenient recharging
5. Implementation of the LTC3400 regulator for stable power supply
6. Compact design for easy handling in warehouse environments
7. Cost-effective solution for small to medium-scale warehouses
8. Operating conditions:
9. Temperature ranges: [-10°C to 40°C]
10. Humidity ranges: [20% to 80%]
11. The Warehouse Barcode Scanner is optimized for indoor use within warehouse settings, ensuring reliable performance under typical operating conditions.



Design Flow

1. **Component Selection:**
    - Identify and select components necessary for the project.
2. **Project Presentation:**
    - Present the project overview, objectives, and expected outcomes.
3. **Schematic Design:**
    - Utilize OrCAD Capture CIS to create the component library and schematic design.
4. **Padstack Design:**
    - Design padstacks for components using OrCAD padStack editor or locate available ones.
5. **Footprint Creation:**
    - Create footprints for components using OrCAD PCB editor or find existing ones.
6. **PCB Layout:**
    - Develop the PCB layout using OrCAD PCB Designer.
7. **Manufacturing and Assembly:**
    - Proceed with manufacturing the PCB, soldering components onto it, and assembling the board.
8. **Firmware Development:**
    - Develop firmware for the Microcontroller to enable functionality.

Each phase is crucial for the successful execution of the project, leading from initial planning to the final implementation and testing.

Block Diagram

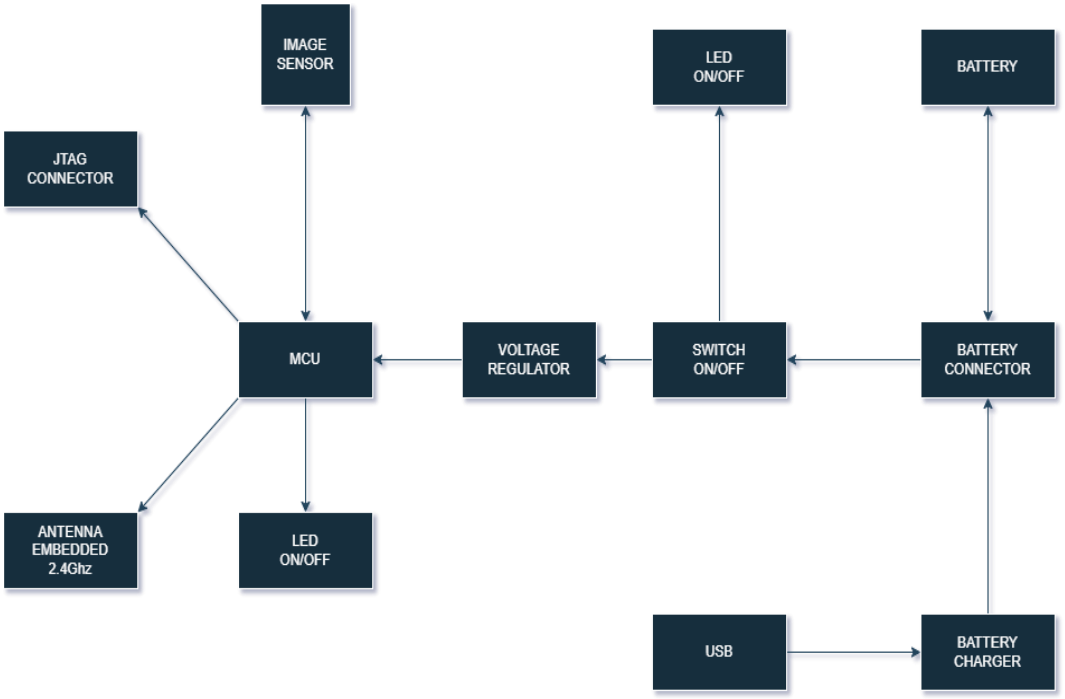


Figure 3.1: Block diagram of the system

The image displayed illustrates the high-level block diagram of our system. The primary functional components are outlined as follows:

Microcontroller: The central processing unit, along with necessary components like decoupling capacitors and quartz crystals.

CMOS Image Sensor: CMOS Image Sensor: Comprising the image sensor itself and essential supporting components such as capacitors and resistors. Since it operates digitally, no specialized conditioning circuitry was required.

LiPo Rechargeable Battery: Serving as the primary power source for the entire system.

USB Connector: Utilized for recharging the battery.

Antenna: A 2.4 GHz patch antenna.

Voltage Regulator: Including both the voltage regulator component and associated circuitry to adjust the battery voltage to the required 3.3V supply.

JTAG Connector: Employed for interfacing the device with a computer and programming it.

ON/OFF Switch: Incorporating the switch itself and its debouncing circuit to facilitate system power control.

Red and Green and blue LEDs: Indicators used to signal the charging status of the device and also the status of power (on/off) and the last one for barcode scanning noticing.

Components

2.1 CMOS Digital Image Sensor: AR0130CS

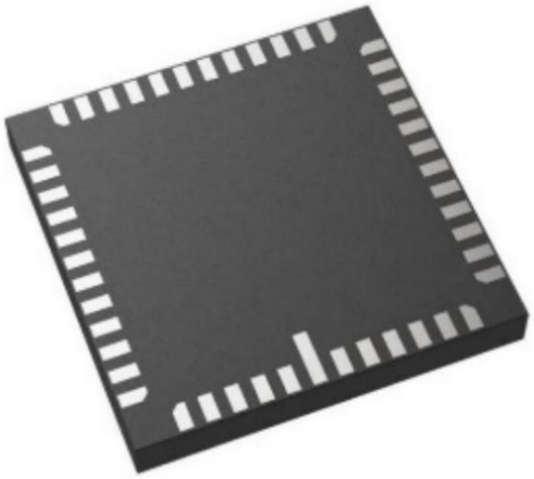
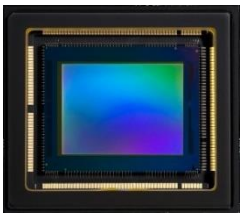


Figure 4.1: CMOS Digital Image Sensor

2.2 Main Features



4 In our endeavor to integrate the Himax HM01B0-AWA image sensor into our project, we encountered difficulties obtaining the precise dimensions necessary to create a footprint for PCB integration. Due to this lack of critical information and the pressing timeline of our project, we made the executive decision to opt for the AR0130CS image sensor as a viable alternative. This choice not only ensures timely progress but also offers compatibility with existing infrastructure and readily available documentation, minimizing potential setbacks

onsemi AR0130 is a 1/3−inch CMOS digital image sensor with an active−pixel array of 1280H x 960V. It captures images with a rolling−shutter readout. It includes sophisticated camera functions such as auto exposure control, windowing, and both video and single frame modes. It is programmable through a simple two−wire serial interface. The AR0130 produces extraordinarily clear, sharp digital pictures, and its ability to capture both continuous video and single frames makes it the perfect choice for a wide range of applications, including our purpose which is scanning the barcodes.
The pin configuration of the sensor and the accompanying conditioning circuitry, which served as the foundation for our system, are depicted in the two images provided below:

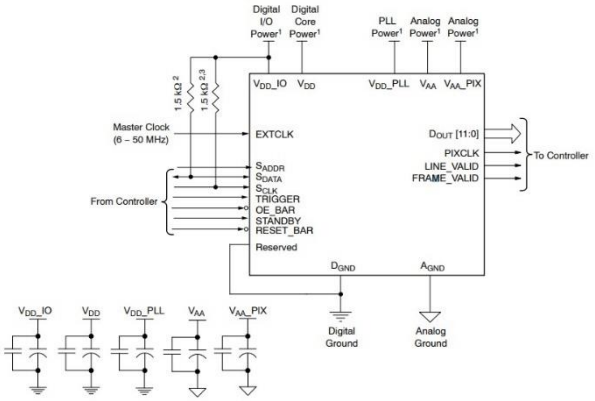


Figure 4.2: Sensor circuitry

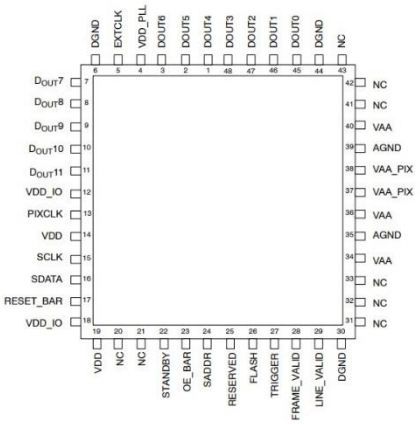


Figure 4.3: Sensor

The main electrical specifications of the sensor are reported below:

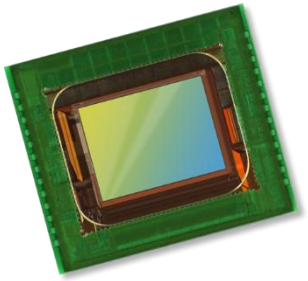
Symbol	Definition	Condition	Min	Typ	Max	Unit
V_{DD}	Core Digital Voltage		1.7	1.8	1.95	V
V_{DD_IO}	I/O Digital Voltage		1.7/2.5	1.8/2.8	1.9/3.1	V
V_{AA}	Analog Voltage		2.5	2.8	3.1	V
V_{AA_PIX}	Pixel Supply Voltage		2.5	2.8	3.1	V
V_{DD_PLL}	PLL Supply Voltage		2.5	2.8	3.1	V
V_{DD_SLVS}	Digital Supply Voltage	Do not connect.	–	–	–	V
V_{IH}	Input HIGH Voltage		V_{DD_IO} * 0.7	–	–	V
V_{IL}	Input LOW Voltage		–	–	V_{DD_IO} * 0.3	V
I_{IN}	Input Leakage Current	No Pull-up Resistor; V_{IN} = V_{DD_IO} or D_{GND}	20	–	–	µA
V_{OH}	Output HIGH Voltage		V_{DD_IO} – 0.3	–	–	V
V_{OL}	Output LOW Voltage		–	–	0.4	V
I_{OH}	Output HIGH Current	At Specified V_{OH}	–22	–	–	mA
I_{OL}	Output LOW Current	At Specified V_{OL}	–	–	22	mA

Figure 4.4: Sensor electrical parameter

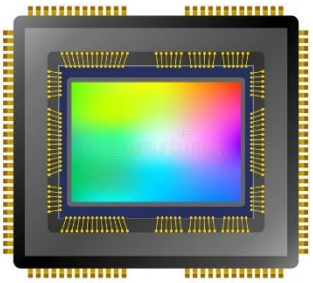


2.3 Alternative Available Choices

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Omnivison - OV07251-G04A-1F



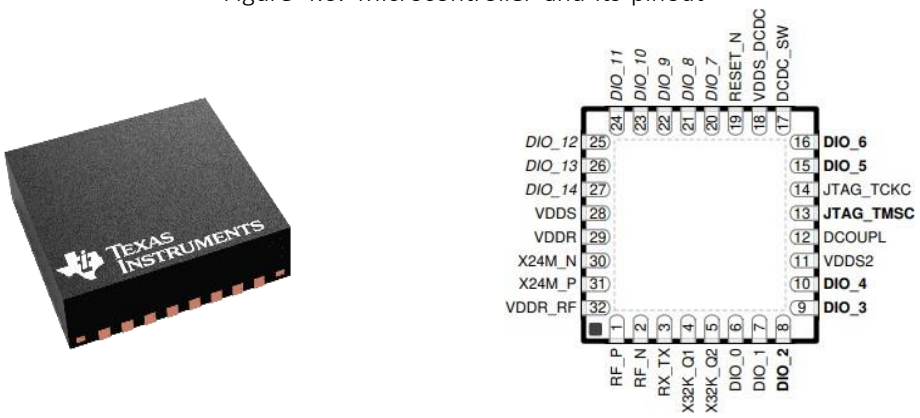
Renesas Electronics - UPD8873CY-A

Figure 4.5: MQ135 on the left and USEQ on the right

Before selecting our first sensor, we evaluated two alternatives: the Omnivision OV07251-G04A-1F and the Renesas Electronics UPD8873CY-A. The Omnivision sensor boasted high-resolution capabilities and promising performance in various lighting conditions, making it a strong contender. However, its higher cost compared to the Himax HM01B0-AWA and potential integration complexities led us to explore other options. The Renesas Electronics sensor offered competitive features, including reliable performance and compatibility with our project requirements. Nonetheless, concerns arose regarding its resolution and suitability for barcode scanning applications. Despite our initial considerations, we ultimately opted for the Himax HM01B0-AWA. However, we encountered unexpected challenges as crucial details, such as the width and length of the sensor, were not provided in its datasheet. This absence of information hindered our ability to create the necessary footprint and integrate the sensor into our PCB design effectively. As a result, we faced significant delays and complications in the development process, prompting us to pivot to the AR0130. The AR0130 emerged as the optimal solution for our barcode scanner project due to its superior resolution, dependable low-light performance, proven reliability, and cost-effectiveness, addressing the shortcomings of both the Himax and Renesas alternatives.

2.4 Microcontroller: CC2640

Figure 4.6: Microcontroller and its pinout



The cc2640 microcontroller boasts a robust ARM Cortex-M3 core, capable of reaching clock speeds of up to 48MHz. Its debugging interface supports either 2 or 4-pin JTAG configurations for flexible debugging options. In terms of communication protocols, it offers support for I2C, UART, and I2S, ensuring versatility in connecting with various peripherals. Notably, it features a range of low power modes, with standby current as low as 1.5 µA and shutdown current at an impressive 100 nA, enhancing energy efficiency.

The microcontroller operates within a supply voltage range of 1.8V to 3.8V, catering to different power supply configurations. Its compatibility with Low Energy Bluetooth technology extends its connectivity options, making it suitable for diverse IoT and wireless applications. In active mode, it consumes moderate power, with RX at 5.9 mA, TX at 0 dBm at 6.1 mA, and TX at +5 dBm at 9.1 mA, balancing performance and energy consumption effects.

2.5 Battery: LP503035



Figure 4.7: LiPo Battery

The LP503035 lithium polymer battery is a compact power solution commonly utilized in small electronic devices. With a nominal capacity of approximately 500mAh and a voltage rating of 3.7 volts, it provides reliable power for various portable gadgets and wearables. Measuring about 37mm x 30mm x 5mm, its compact size is advantageous for space-constrained applications. Moreover, it operates efficiently within a temperature range of -20°C to 60°C (-4°F to 140°F), ensuring stable performance across diverse environmental conditions.

2.6 Battery charger: MCP73833

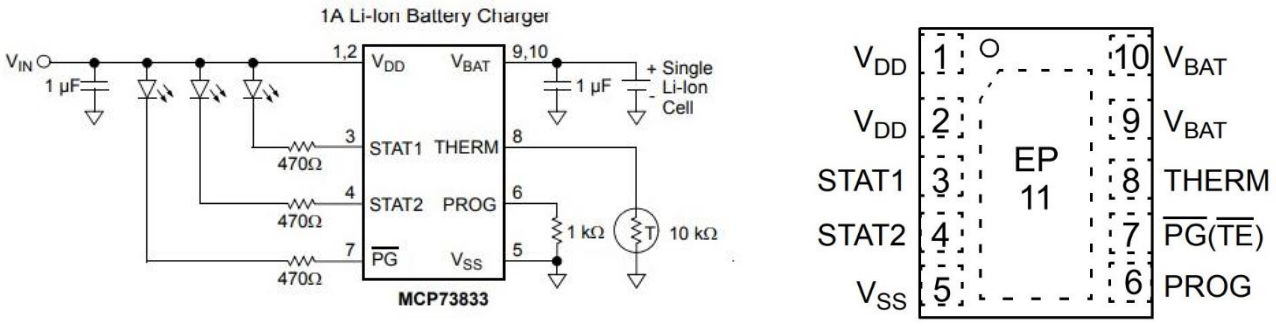


Figure 4.8: Battery charger scheme and package

Pin No.		Symbol	Function
DFN-10	MSOP-10		
1	1	V_{DD}	Battery Management Input Supply
2	2	V_{DD}	Battery Management Input Supply
3	3	STAT1	Charge Status Output
4	4	STAT2	Charge Status Output
5	5	V_{SS}	Battery Management 0V Reference
6	6	PROG	Current Regulation Set and Charge Control Enable
7	7	$\overline{PG}$, $\overline{TE}$	**MCP73833:** Power Good output, **MCP73834**: Timer Enable input
8	8	THERM	Thermistor input
9	9	V_{BAT}	Battery Charge Control Output
10	10	V_{BAT}	Battery Charge Control Output
11	—	EP	Exposed Thermal Pad (EP); must be connected to V_{SS}.

Figure 4.9: Battery charger pinout

The MCP73833 is a highly integrated linear charge management controller for lithium-ion and lithium-polymer batteries. It offers autonomous charging and includes features like constant current and constant voltage regulation, as well as thermal regulation for safety. With a compact form factor and low external component count, it's widely used in portable electronic devices for efficient battery charging.
key specifications:
Input Voltage Range: Typically, 4.5V to 6.5V.
Charge Current: Adjustable up to 500mA.
Temperature Range: -40°C to +85°C.
Charge Termination: Charge termination by voltage and current for safety.
Preconditioning: Automatic preconditioning to recharge deeply discharged cells.
Automatic Recharge: Automatically restarts charging if battery voltage falls below an internal threshold.

2.7 Voltage Regulator: LTC3400

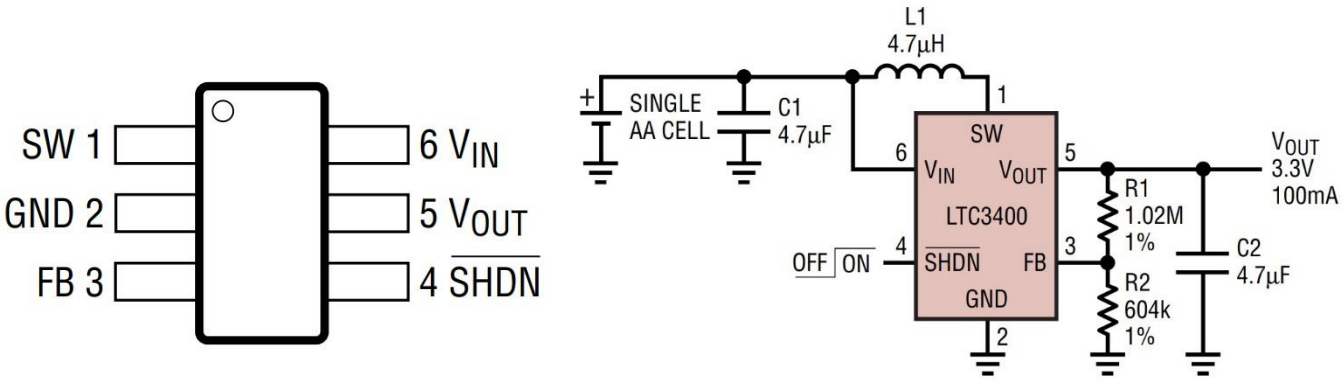


Figure 4.10: Voltage regulator schemes

PIN	NAME	FUNCTION
1	POK	Open-Drain Power-OK Output. POK is high impedance when output voltage is in regulation. POK sinks current when V_{FB} falls below 1.1V. Connect a 10kΩ to 1MΩ pull-up resistor from POK to V_{OUT} for a logic signal. Ground POK or leave unconnected if not used. POK is high impedance in shutdown.
2	$\overline{SHDN}$	Shutdown Input. Drive high for normal operation; drive low for shutdown mode. OUT is high impedance in shutdown.
3, 4	IN	Input Supply. Connect both pins together and bypass to GND with a ceramic capacitor (see *Capacitor Selection* section).
5	GND	Ground. Connect GND to PGND with a short trace.
6	PGND	Power Ground. Charge-pump current flows through this pin.
7	CXN	Negative Terminal of the Charge-Pump Transfer Capacitor
8	CXP	Positive Terminal of the Charge-Pump Transfer Capacitor
9	OUT	Power Output. Bypass to GND with an output filter capacitor.
10	FB	Dual-Mode Feedback. Connect FB to GND for 3.3V output. Connect to an external resistor divider to adjust the output voltage from 2.5V to 5.5V.

Figure 4.11: Voltage regulator pinout

Buck/boost regulating charge pump used to adapt and stabilize the battery output voltage to 3.3V. Its main features are listed below:

Extended operating voltage range: 2.5V to 5.5V
Fixed or adjustable output voltage: 3.3V or adjustable down to 0.8V
Temperature range: -40°C to 125°C
10-pin µMAX package

Up to 600 mA output current.

2.8 Antenna

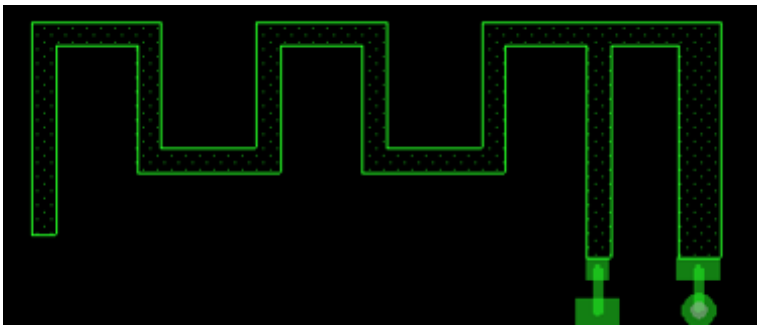
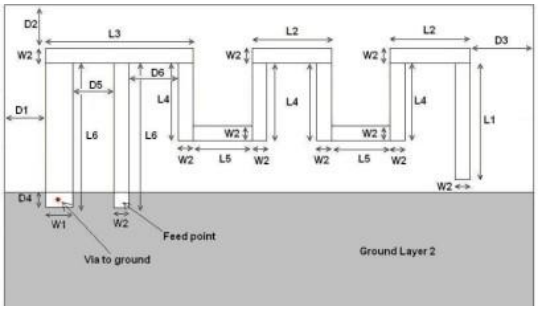


Figure 2.8.1: Antenna prototype



L1	3.94 mm
L2	2.70 mm
L3	5.00 mm
L4	2.64 mm
L5	2.00 mm
L6	4.90 mm
W1	0.90 mm
W2	0.50 mm
D1	0.50 mm
D2	0.30 mm
D3	0.30 mm
D4	0.50 mm
D5	1.40mm
D6	1.70 mm

Figure 2.8.9: Antenna dimensions

The chosen antenna is designed to operate at 2.4 GHz with a characteristic impedance of 50 Ω. While an antenna component was considered as an alternative solution, the decision was made to utilize the printed version for this project.

2.9 USB Connector

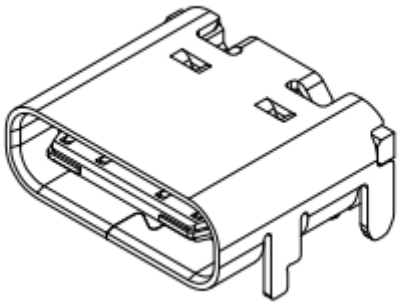


Figure 2.9.1: USB connector model

The Type C USB connector facilitates the recharging of the battery using an external USB cable.

2.10 led

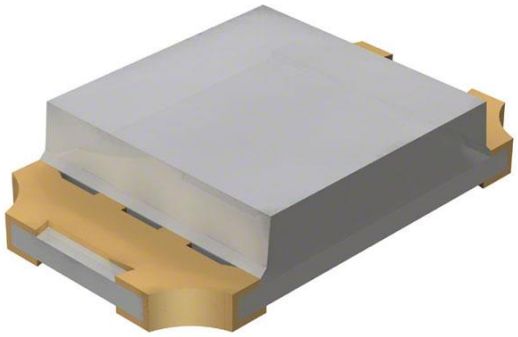


Figure 2.10.1: RGB led design and model

We are using TWO LEDs to announce to the user whenever the barcode has been scanned and also whenever for the device is on or off.

Regarding the RGB LEDs, their operation can be specified during the Microcontroller firmware programming phase. In our application, they will indicate the battery status, signalling whether it is discharging or currently charging. We intend to program them so that they emit a blue light while charging, switch to red when the battery charge is low, and turn green once fully charged.
Additionally, we've integrated two other LEDs for specific purposes. One will notify the user whenever a barcode is scanned (indicated by a green light), while the other will indicate whether the device is powered on or off (shown by a red light)

2.11 Mechanical Switch

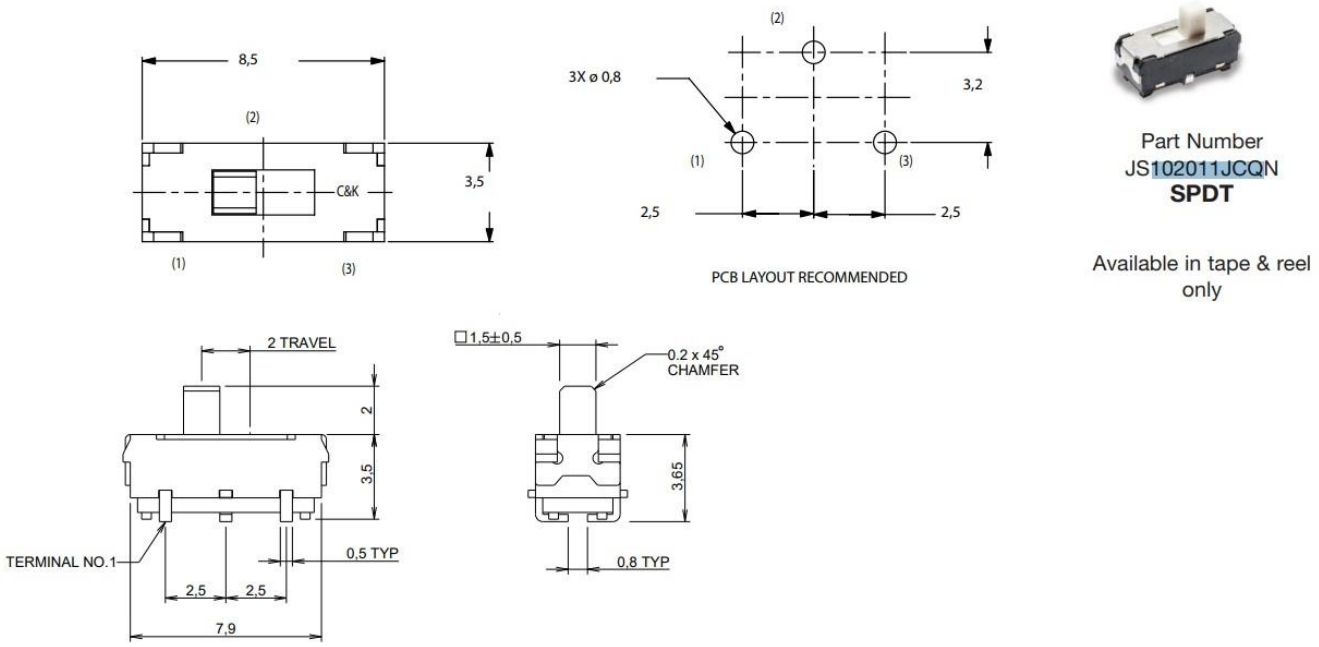


Figure 4.18: Green and Red electrical parameters

The switch serves the purpose of activating and deactivating the entire system. Being one of the limited through-hole components, it necessitated the inclusion of through-hole pads.

2.12 JTAG Connector

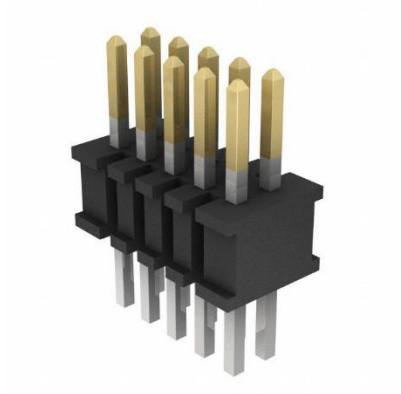


Figure 4.19: Jtag connector

The FSTH 105 01 L D is a 4x2 2.54 mm SMD pin header connector. It is utilized in our design as a JTAG connector, facilitating debugging and programming of embedded systems. Additionally, it features a robust design, ensuring reliable connections in demanding environments

Power consumption estimations

Main power consumption contribution:

Microcontroller: Active-Mode 1,45mA+31µA/MHz, Stand-by-Mode 1µA;
Image sensor: Operating Mode 15mA, Standby mode 200uA;
LED: Active Mode 0.5mA;

Taking a measurement every 10 seconds we have on average:

Image sensor

$$\frac{15uA*1+2couA*9}{10} = 1.52mA$$

Microcontroller:

$$\frac{1uA * 9s \text{ (standby mode)} + 9.1mA * 1s + (1.45+ 0.31 * 48) * 1s \text{ (active mode)}}{10} = 0.63mA$$

LED (to scan barcodes)

$$\frac{1mA* 1s+0.01mA*9s}{10} = 0.19mA$$

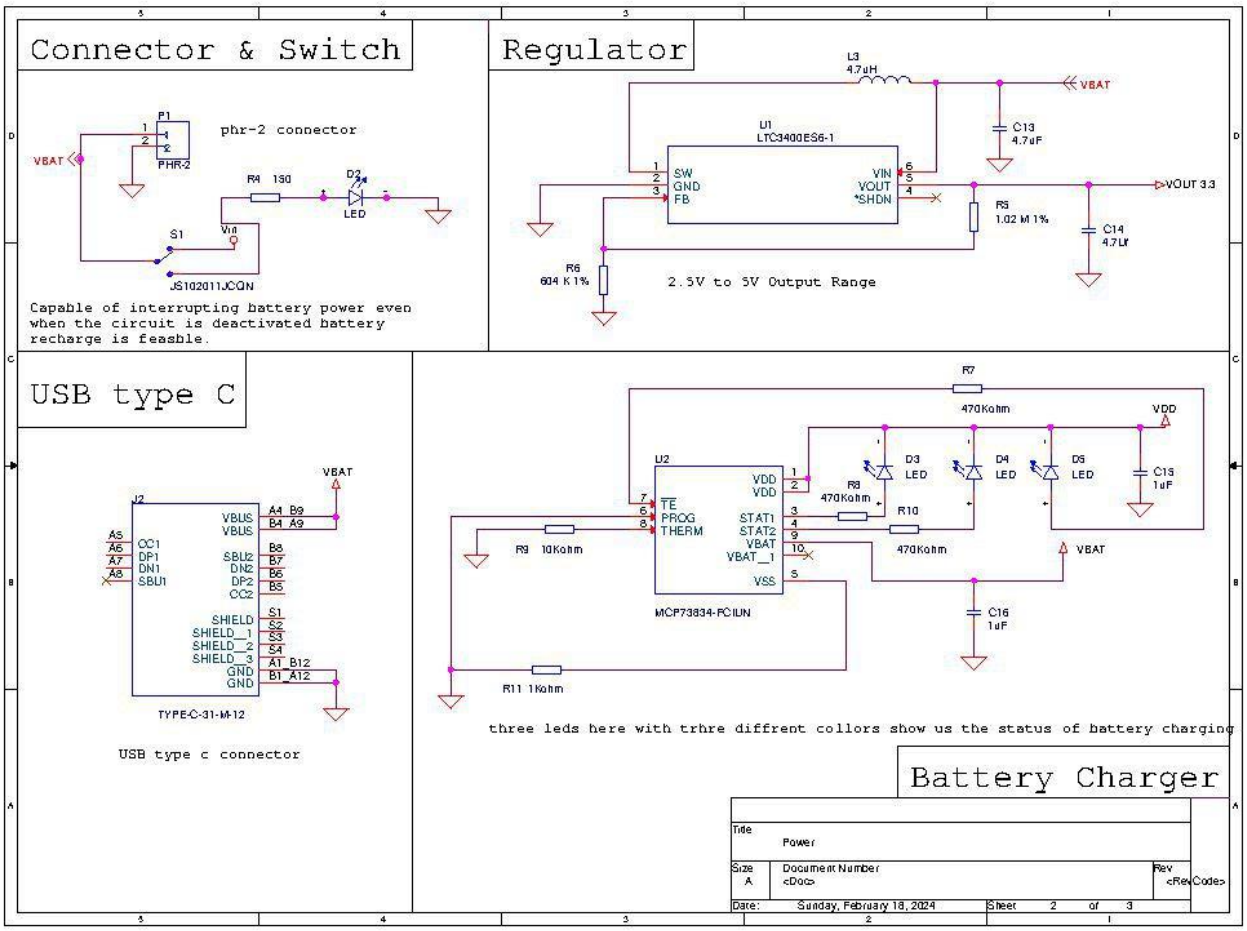
LED (power LED)

1mAh

Total battery duration :

$$\frac{500mAh}{1.52mA+1.63mA+0.19mA+10mA} = 37\ h$$

Schematic



4.1 Power section

5 Voltage regulator, battery charger, switch, connector and usb port schematic have been implemented following datasheet suggestions.

4.2 MicroController section

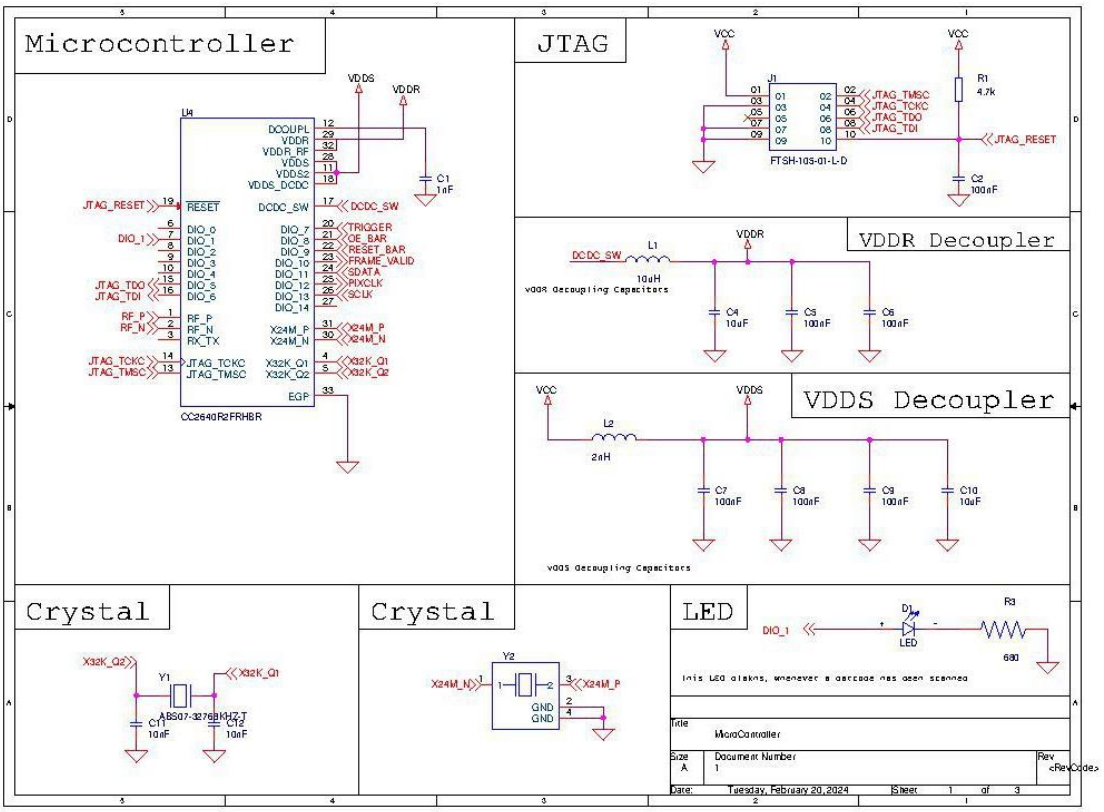


Figure 5.4: Microcontroller schematic

Antenna is configured for single ended operations. All the passive elements have been selected according to datasheet specifications. No capacitors have been used on the 24MHz quartz because they are already included in the Microcontroller. VDDR and VDDR RF require a 1.7-1.95V supply voltage that can be generated from DCDC Microcontroller internal regulator.

4.3 Sensor section

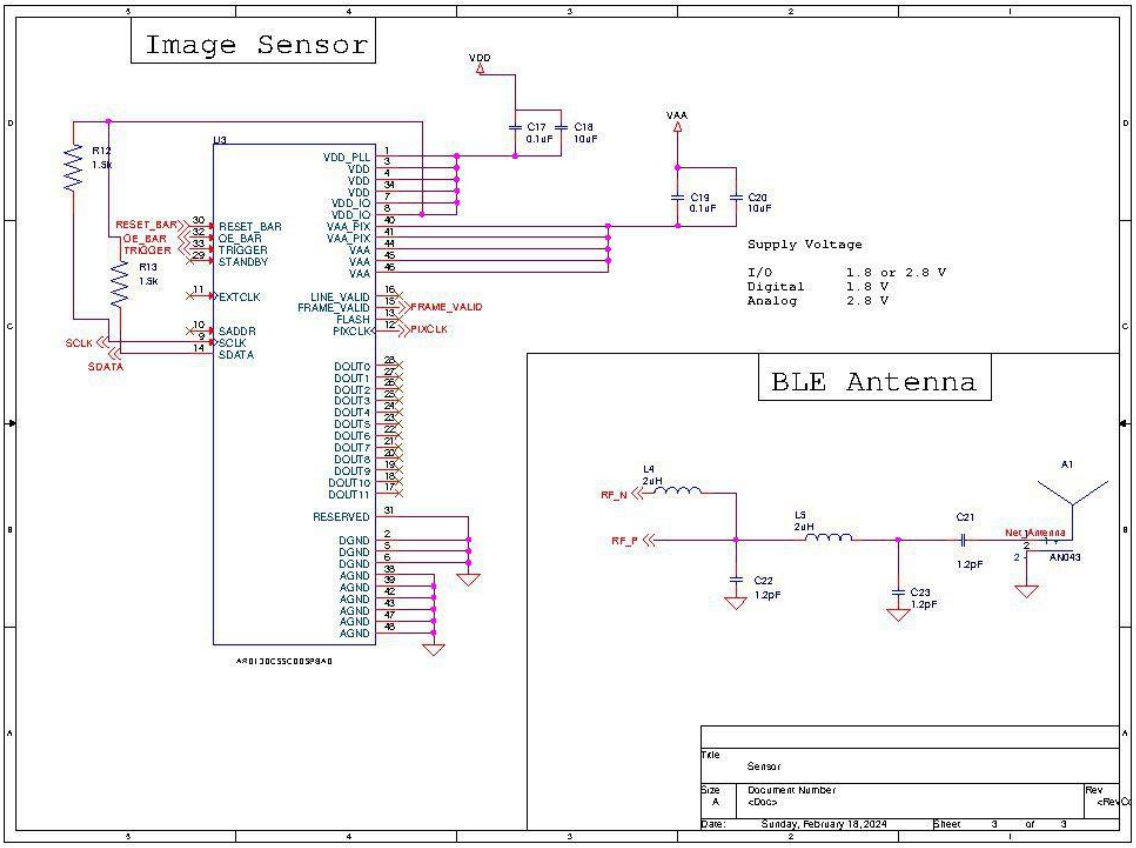


Figure 5.5: Sensor schematic

Sensor scheme has been realized according to datasheet specifications. 2 10kΩ resistors and 2 capicitors respectively 0.1 uF and 10uF have beenused as pull-up for VDD and VAA.

PCB Layout

The initial phase of PCB design focused on developing the top layer, with careful consideration given to the placement of components. Priority was given to minimizing the distance between the microcontroller and antenna to prevent potential disruptions. Additionally, a sensor was strategically positioned along the board edge to ensure precise measurements. Incorporating a switch allowed for convenient circuit disconnection from the battery while keeping the battery charger operational for recharging purposes. Notably, the routing process presented a notable challenge, particularly with the need to accommodate seven wires from the sensor to the microcontroller solely on the top layer. Our objective to avoid using vias for signal transmission added complexity to the task. However, we successfully overcame this hurdle, achieving seamless connectivity without compromising signal integrity. It's worth mentioning that vias were exclusively utilized for grounding signals, further enhancing the overall performance and reliability of the design. This meticulous approach to component placement and routing contributed to the board's Class 7 classification by Eurocircuit standards.

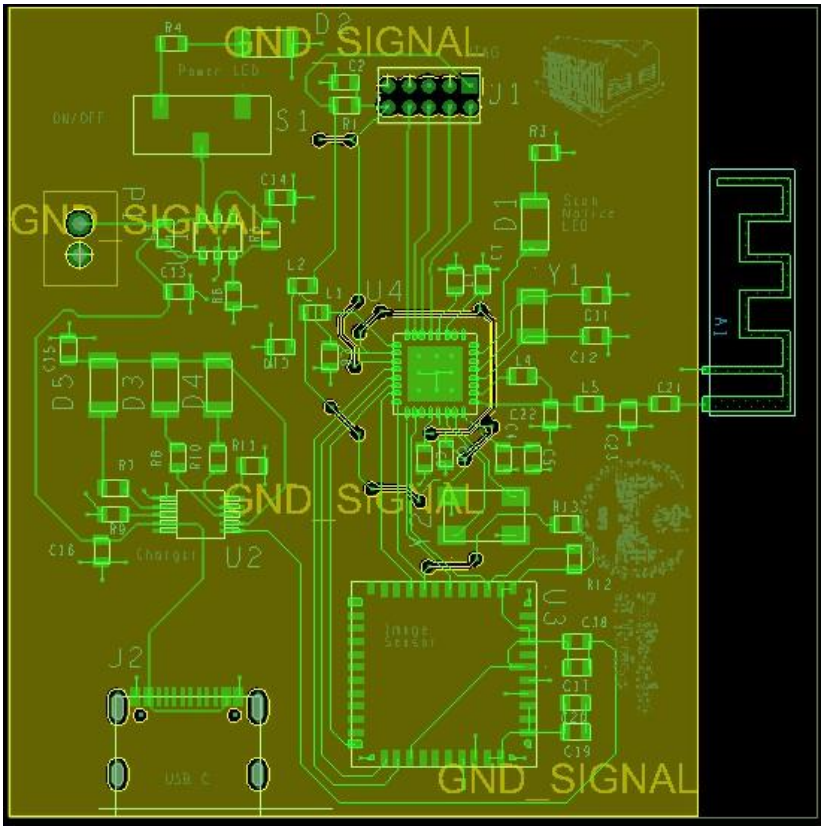


Figure 6.1: PCB

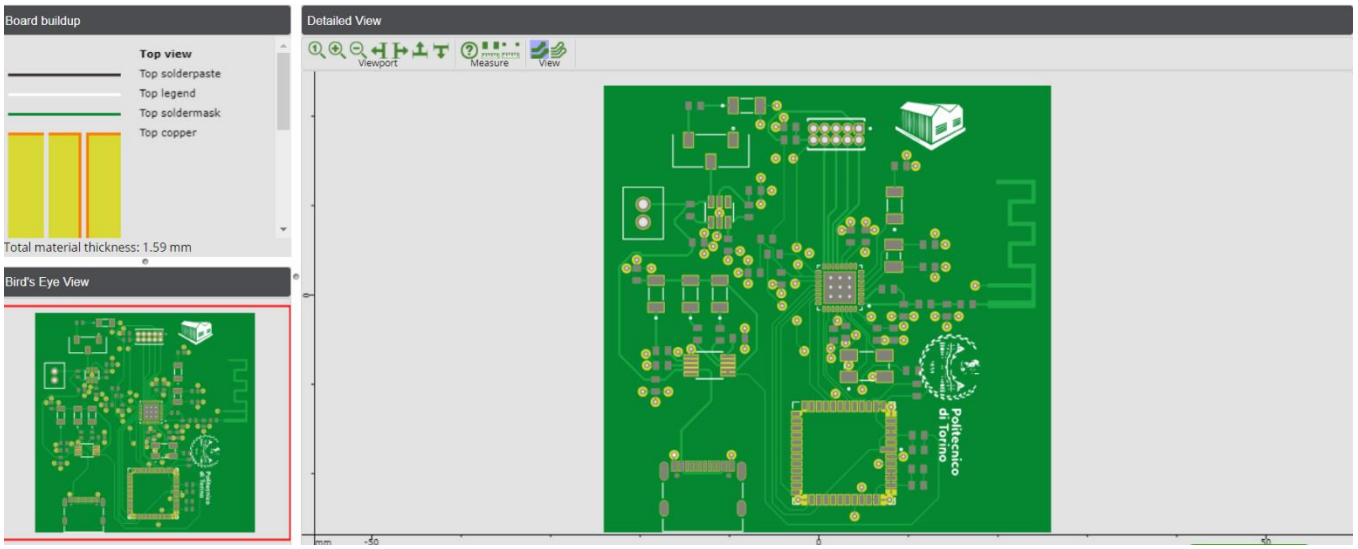


Figure 6.2: PCB top

Main component of the bottom layer is the ground plane. It has been designed looking for the minimal number of interruptions. This is important to let return current flow in a straight path,avoiding in this way spurious EMC emission.

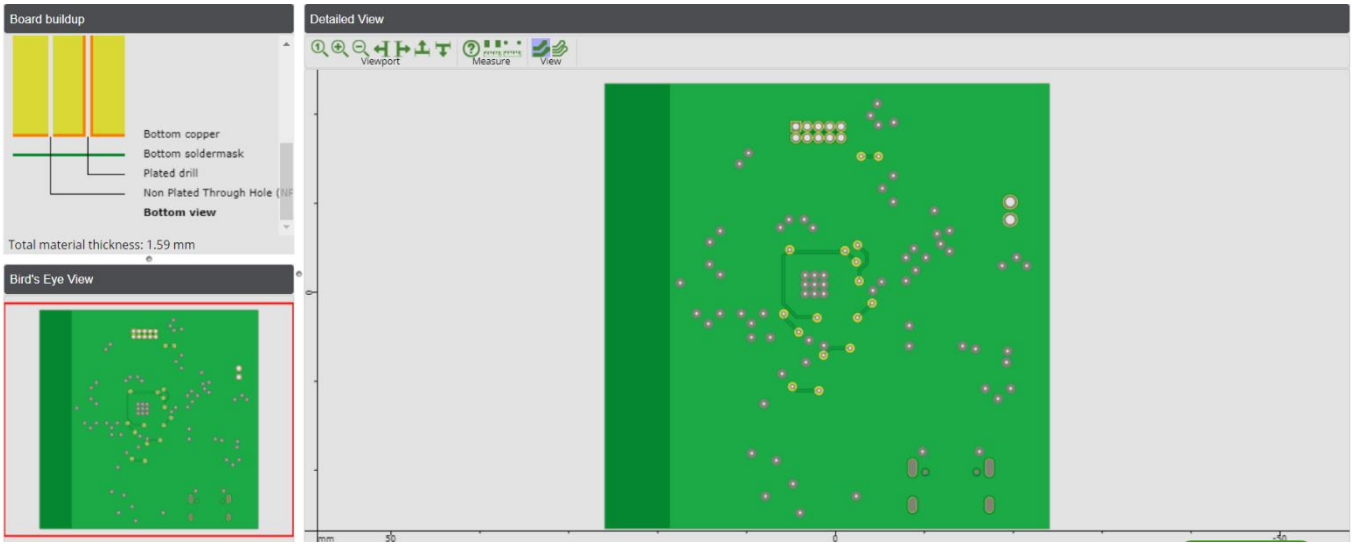


Figure 6.3: PCB bottom

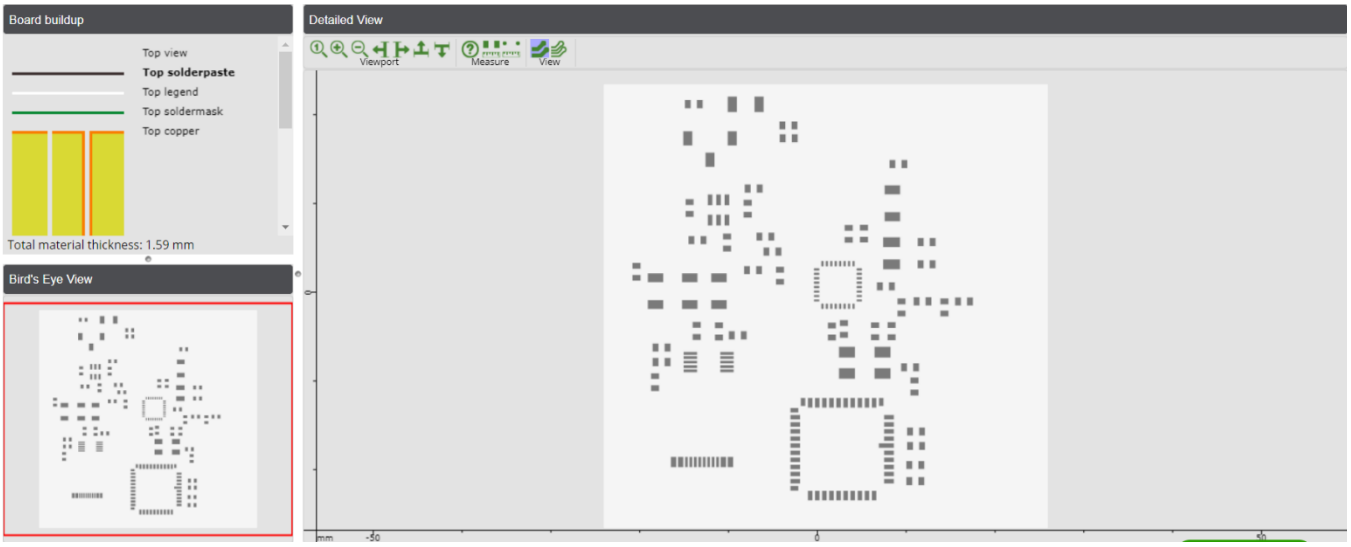


Figure 6.4: PCB top solderpaste

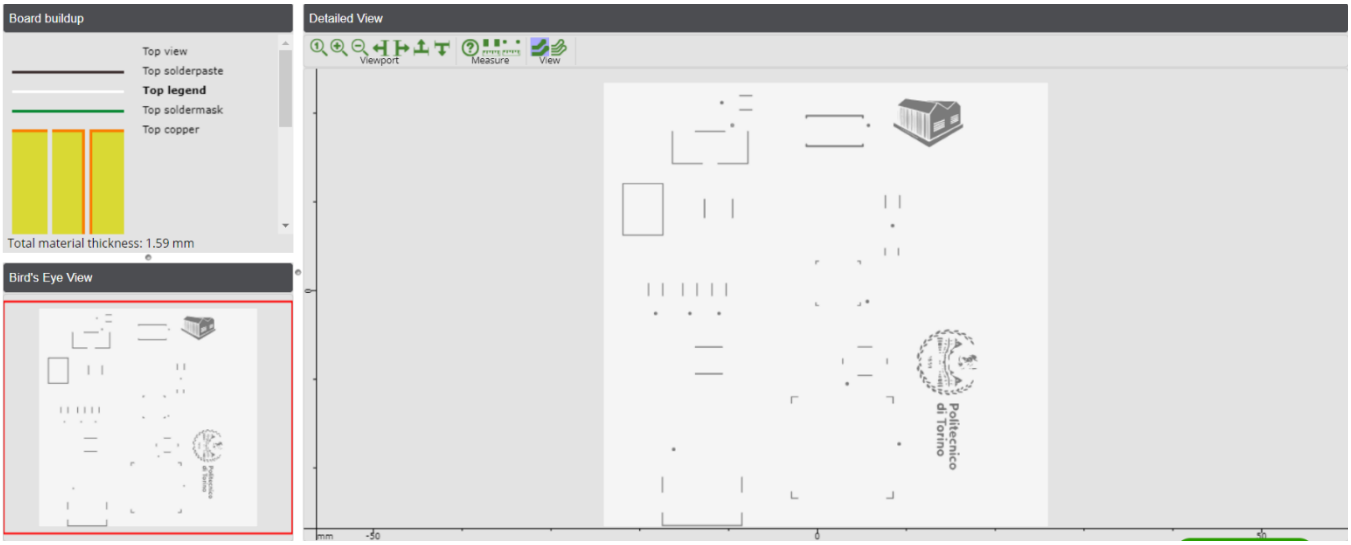


Figure 6.5: PCB top legend

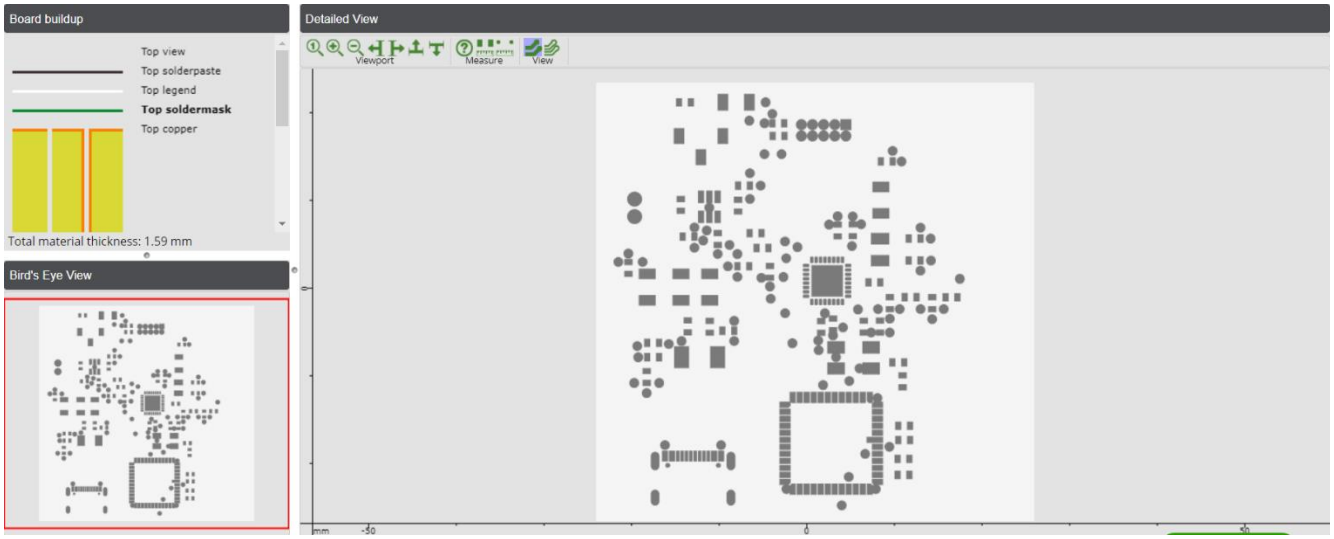


Figure 6.6: PCB Top soldermask

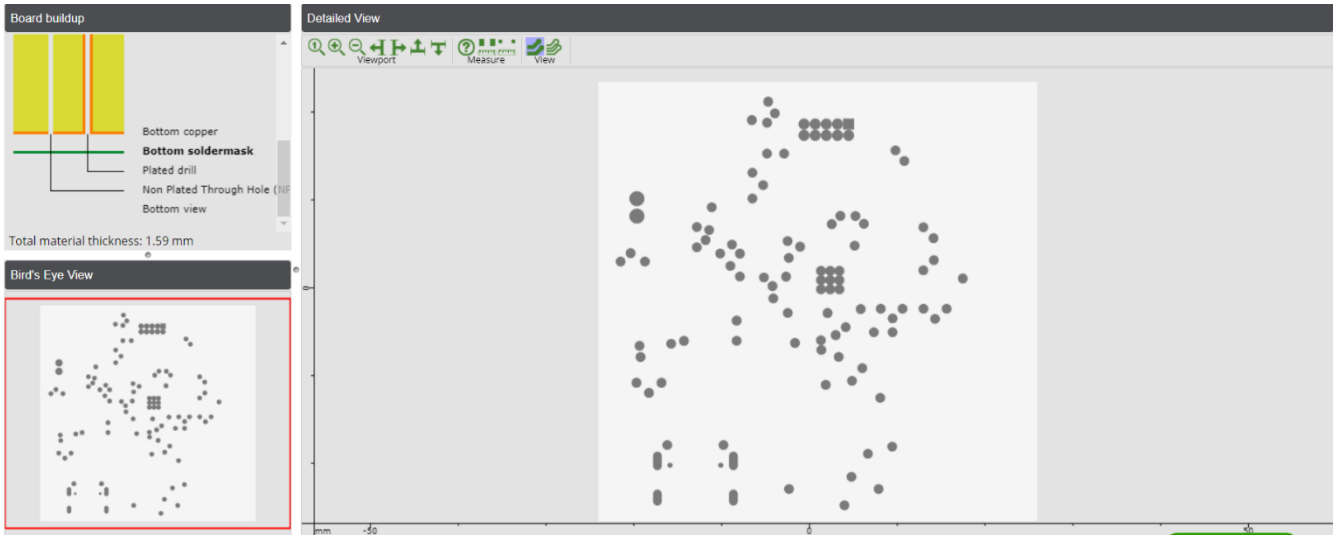


Figure 6.7: PCB Bottom soldermask

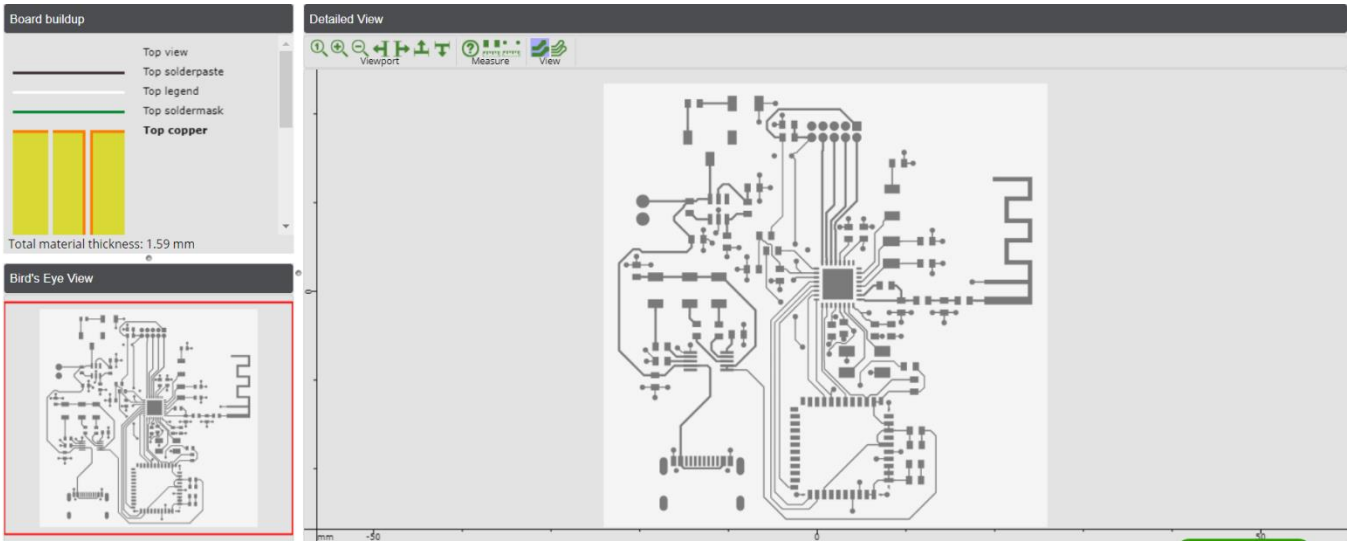


Figure 6.8: Top cooper

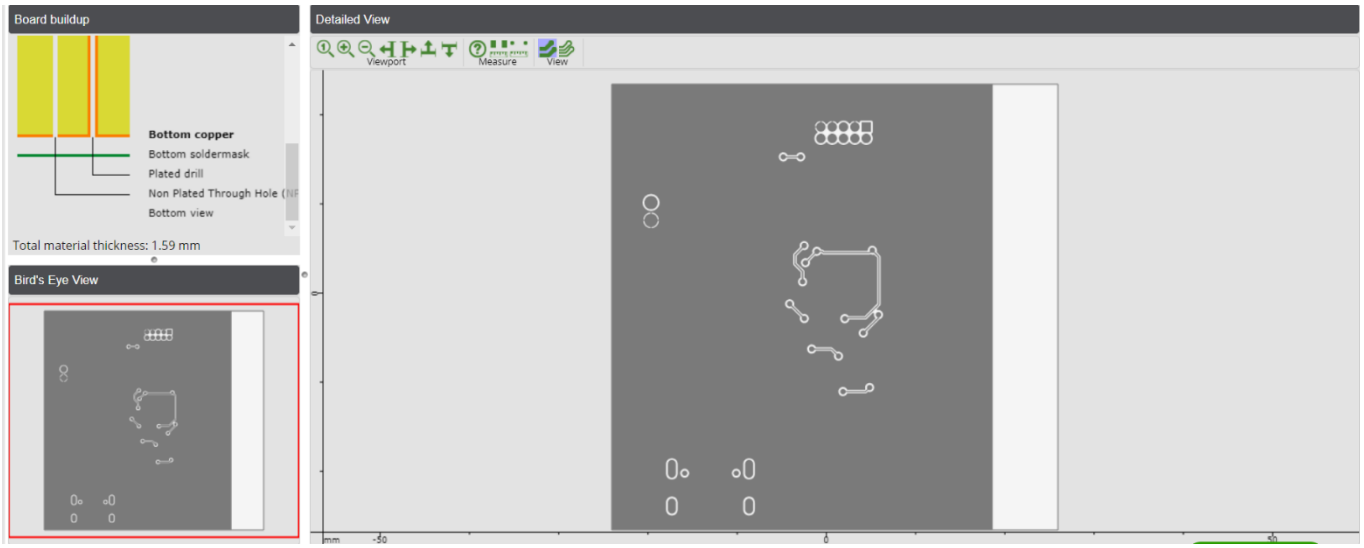
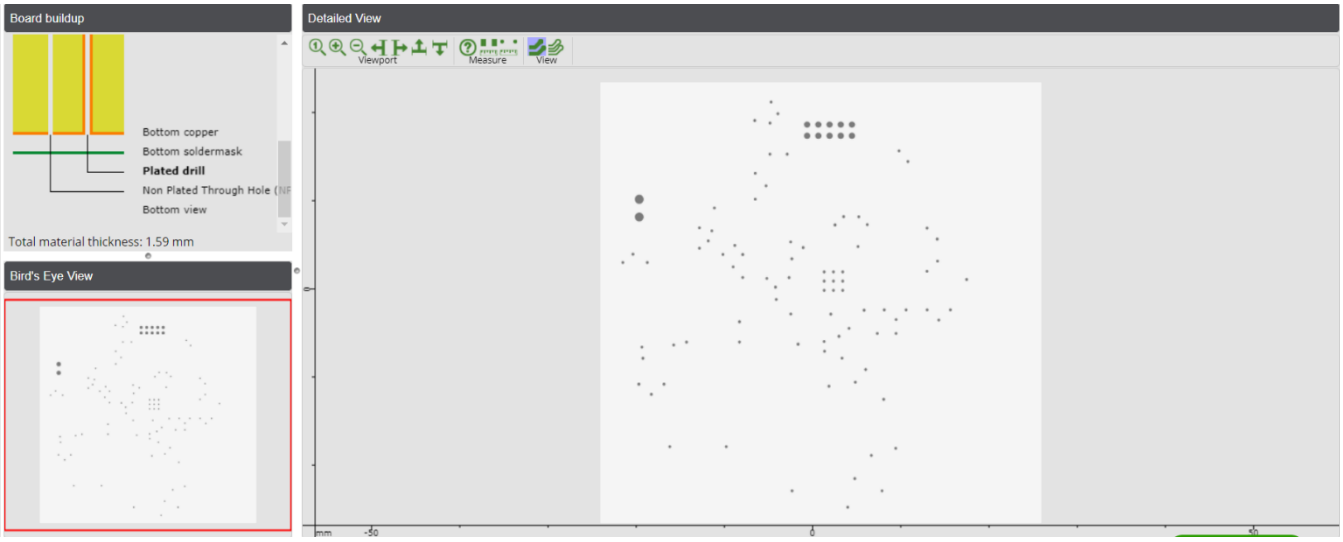


Figure 6.9: Bottom cooper



igure 6.10: Plated Dril

Bills of Materials

The provided Bill of Materials (BOM) is customized for a particular manufacturing instance, focusing on producing a single PCB. Yet, to assess costs more comprehensively, estimates for 50, 100, and 1000 unit productions are included.

Item	Quantity Per Board	Manufacturer Part #	Reference Designators	website	website Part Number	Value	Mounting type	Part Description	Manufacturer	UNIT PRICE in euro
1	1	YC0002AA	A1	/	277-YC0002AA	50 Ohms	/	50 Ohm Antenna	Quectel	1,15
2	1	CDR32BX103BKUS	C12	Mouser	581-CDR32BX103BKUS	10nF	SMD	Multilayer Ceramic Capacitors MLCC - SMD/SMT .01UF 100V 10%	KYOCERA AVX	4,76
3	1	KGM15ACG1H102FT	C1	Mouser	581-KGM15ACG1H102FT	1nF	SMD	Multilayer Ceramic Capacitors MLCC - SMD/SMT 50V .001uF C0G 0603 1%	KYOCERA AVX	0,214
4	3	0603N1R2C500CT	C22,C21,C23	Mouser	791-0603N1R2C500CT	1.2pF	SMD	MLCC - SMD/SMT 1.2pF +-0.25% 50V	Walsin	0,108
5	2	CL10B105KA8NNND	C15,C16	Mouser	187-CL10B105KA8NNND	1uF	SMD	MLCC - SMD/SMT 1uF+/-10% 25V X7R 16 0603	Samsung Electro-Mechanics	0,093
6	7	CL10B104KA8NNNC	C2,C5,C6,C8,C9,C17,C19	Mouser	187-CL10B104KA8NNNC	100nF	SMD	MLCC - SMD/SMT 100nF+/-10% 25V X7R 1608	Samsung Electro-Mechanics	0,093
7	4	CL10A106MA8NRNC	C20,C18,C10,C4	Mouser	187-CL10A106MA8NRNC	10uF	SMD	MLCC - SMD/SMT 10uF+/-20% 25V X5R 1 0603	Samsung Electro-Mechanics	0,233
8	2	CL10A475MA8NQNC	c13, c14	Mouser	187-CL10A475MA8NQNC	4.7uF	SMD	MLCC - SMD/SMT 4.7uF+/-20% 25V X5R 0603	Samsung Electro-Mechanics	0,093
9	1	CSL1001BT1C	D4	Mouser	755-CSL1001BT1C	LED B	SMD	Standard LEDs - SMD LOW CURRENT BLUE SMD LED	ROHM Semiconductor	0,512
10	2	SML-P11MTT86R	D2,D5	Mouser	755-SML-P11MTT86R	LED G	SMD	Standard LEDs - SMD ULTRA COMPACT YELLOW-GREEN	ROHM Semiconductor	0,391
11	2	SML-P11VTT86R	D1, D3	Mouser	755-SML-P11VTT86R	LED R	SMD	Standard LEDs - SMD MINI-MOLD ULTRA COMPACT RED	ROHM Semiconductor	0,502
12	1	LQM18DZ100M70L	L1	Mouser	81-LQM18DZ100M70L	10uH	SMD	Power Inductors - SMD 0603 10uH 20% 300mA 1.37ohm AEC-Q200 Simulate & compare characteristics in SimSurfing Simulate & compare characteristics in	Murata Electronics	0,205
13	1	BLM18HE152SN1D	L2	Mouser	81-BLM18HE152SN1D	2nH	SMD	Ferrite Beads 1500 OHM 25% Alternate Sizing Guide Below Simulate & compare characteristics in SimSurfing Simulate & compare characteristics in SimSurfing	Murata Electronics	0,177
14	1	SRP5030CC-4R7M	L3	Mouser	652-SRP5030CC-4R7M	4.7uH	SMD	Power Inductors - SMD Ind,5.7x5.2x2.8mm,4.7uH+/-20%,4A,shd	Bourns	0,772
15	2	LQW21FT2R0M0HL	L4,L5	Mouser	81-LQW21FT2R0M0HL	2uH	SMD	RF Inductors - SMD 0805 2.uH 20% 450mA 0.29ohm AEC-Q200	Murata Electronics	1,36
16	1	CMP2512-FX-1001ELF	R11	Mouser	652-CMP2512-FX-1001L	1k	SMD	Thick Film Resistors 1K OHM 1%	Bourns	0,456
17	3	CR0603-FX-1501ELF	R12,R3,R4	Mouser	652-CR0603FX-1501ELF	1.5k	SMD	Thick Film Resistors - SMD 1.5K 1% 1/10W	Bourns	0,093
18	1	ERJ-1GJF6800C	R3	Mouser	667-ERJ-1GJF6800C	680	SMD	Thick Film Resistors - SMD 0201 0.05W 1% 680Ohm AEC-Q200	Panasonic	0,177
19	1	RCL12184K70FKEK	R1	Mouser	71-RCL12184K70FKEK	4.7k	SMD	Thick Film Resistors - SMD 1watt 4.7Kohms 1% 1218 100ppm	Vishay / Draloric	0,642
20	3	CMP1206AFX-4703ELF	R7,R8,R10	Mouser	652-CMP1206AFX-4703L	470k	SMD	Thick Film Resistors - SMD ResHighPowerA 1206 470k 1% 3/4W TC100	Bourns	0,372
21	1	CR0805-FX-6043ELF	R6	Mouser	652-CR0805FX-6043ELF	604k	SMD	Thick Film Resistors 604K 1%	Bourns	0,093
22	1	SMV1W1M2JT	R5	Mouser	279-SMV1W1M2JT	1.02M	SMD	Thick Film Resistors - SMD SMV 1W 1M2 5% SMD Resistor	TE Connectivity / AMP	0,549
23		PHR-2	P1	digikey	455-1165-ND	connector	Free Hanging (In-Line)	CONN RCPT HSG 2POS 2.00MM. 2 Rectangular Connectors - Housings Receptacle Natural 0.079" (2.00mm)	JST Sales America Inc.	0,09000
24	1	JS102011JCQN	S1	Mouser	611-JS102011JCQN	Switch	SMD	Slide Switches .3A 6VDC SPDT VERT MNT SMT J BEND	C&K	0,846
25	1	AR0130CSSM00SPCA0-DPBR	U3	digikey	56-AR0130CSSM00SPCA0-DPBR-I	SENSOR AR0130C	SMD	Optical Sensors - Image Sensors 1.2 MP 1/3 CIS RGB Parallel, BBAR glass	onsemi	£20.60
26	1	CC2640R2FRHBR	U4	Mouser	595-CC2640R2FRHBR	CC2640R2FRHBR	SMD	RF Microcontrollers - MCU SimpleLink Bluetooth low enrgy Wirels MCU	Texas Instruments	4,45
27	1	TYPE-C-31-M-12	J2	lcsc	C165948	USB	SMD	5A 1.16P Female Type-C SMD USB Connectors ROHS	Korean Hroparts Elec	£0.16
28	1	LTC3400ES6-1#TRMPBF	U1	Mouser	584-C3400ES6-1TRMPBF	C3400ES6-1#TRMF	SMD	Switching Voltage Regulators 600mA, 1.2MHz Micropower Synchronous Boost Converter in ThinSO	Analog Devices	5,87
29	1	ABS07-32.768K-T	Y1	Mouser	815-ABS07-32.768K-T	32.768KHz	SMD	Crystals 32.768 KHZ 12.5PF SMD	ABRACON	0,577
30	1	ABM3B-24.000MHZ-B2-T	Y2	digikey	535-9126-2-ND	24MHz	SMD	CRYSTAL 24.0000MHZ 18PF SMD- 24 MHz ±20ppm 18pF 4-SMD crystals, lead-free	Abracon LLC	£10.40
31	1	503035	/	Amazon	B07S84SBV3	BATTERY	/	7V 500mAh 503035 Lipo Battery Rechargeable Lithium Polymer ion Battery Pack with PH 2.0mm JS	AKZYTUE	£8.34
32	1	MCP73834-FCI/UN	U2	Mouser	579-MCP73834-FCI/UN	ATTERY CHARGE	SMD	Battery Management Charge Management Controllers	Microchip Technology	1,06
33	1	FTSH-105-01-L-D	J1	Mouser	200-FTSH10501LD	JTAG	Through Hole	Headers & Wire Housings High Reliability Header Strips, .050" pitch	Samtec	1,96
										£39.50

Figure 8.1: BOM 1 piece

The selected components are predominantly accessible through Mouser, Amazon (for the battery), and LCSC (for the USB connector).

6.1 Cost Estimation

6.1.1 1 PCB

Component	Cost
PCB	52.50e
Components	39.50e
Stencil	31.56e
Discount on eurocircuit	34.40e
Total	89.16e

6.1.2 50 PCB

Component	Cost
PCB	277.53e
Components	941.00e
Stencil	31.56e
Discount on eurocircuit	52.06e
Total	1198.03

6.1.3 100 PCB

Component	Cost
PCB	324.52e
Components	1667.00e
Stencil	31.56e
Discount on eurocircuit	66.60
Total	1956.48e

6.1.4 1000 PCB

Component	Cost
PCB	1378.60e
Components	15790.00e
Stencil	31.56e
Discount on eurocircuit	85.40e
Total	17114.76e

References

Components datasheets:

7.1 Mouser.it

7.2 Lcsc.com

7.3 digikey.it

PCB render and quotations: eurocircuit.com

Technical report and presentations examples:

7.4 Technical report.pdf

7.5 NOxSensor.pdf

7.6 ESE_presentation_NOxSensor_Pecorara_Cora_301179_296164.pdf

7.7 MOVIE_PECORARA_CORA_NOxSensor

Design History of the Project

8 Capture CIS by Cadeence

Upon finalizing the component selection process, we utilized Capture CIS by Cadence to develop our schematic. This enabled us to precisely articulate the intended wiring configuration of our system, ensuring technical clarity and alignment with our design objectives.

8.1 Cadence PCB Padstack Editor

Padstacks were either generated internally or sourced from reputable providers, adhering to Eurocircuits' guidelines to ensure consistent pad sizes and solder mask dimensions.

8.2 Cadence Allegro

Version 1.1

We commenced component placement with the implementation of our antenna and microcontroller. Subsequently, upon integrating crucial files, we thoroughly evaluated alternative configurations for the placement of additional components such as sensors, battery chargers, LEDs, and others.

Version 1.2

We initiated the routing phase; however, encountered difficulties in routing signals on the top board of our PCB. After adjusting the input and output pins of our microcontroller, we successfully routed all signals on the top layer of our board.

Version 1.3

The final step involved routing the bottom layer of our board to accommodate GND signals and incorporating logos onto our board.

